CPU Scheduling Algorithms

LumoraAI Presentation Demo • 02 Operating Systems

Scheduling objectives

• Important scheduling metrics include waiting time, turnaround time, response time and throughput. Different algorithms optimize different objectives and can therefore produce different trade-offs.

Common algorithms

• First-Come, First-Served executes processes in arrival order and can suffer from convoy effects. Shortest Job First minimizes average waiting time under ideal knowledge of job lengths but requires burst-length estimation in practice.

• Round Robin assigns each ready process a time quantum and cycles through the queue. A very small quantum increases context-switch overhead, while a very large quantum makes the policy behave more like FCFS.

Priority scheduling

• Priority scheduling selects a highest-priority ready process. Aging can gradually increase the priority of long-waiting processes to reduce starvation.

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